Type MS · Explain how the placenta, cord and hormones support a fetus

Answer 1 of 4 · Question 8 part b.i · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
8	(a)	(i)	triggers ovulation/causes release of the egg;		1
		(ii)	Three from: • FSH stimulates growth/production of follicles in the ovary; • more eggs mature/produced; • increases the chances of more than one egg being released; • each egg is fertilised by a sperm;		3
	(b)	(i)	Three from: • placenta supplies glucose/oxygen from the mother; • twins have to share the supply of glucose/receive less glucose/oxygen; • less aerobic respiration; • less energy is released in the foetus and less growth occurs;	Allow nutrients for mp's 1 and 2	3
		(ii)	Two from • same egg and sperm used; • zygote splits/divides • forming 2 embryos;		2
	(c)	(i)	75th		1
		(ii)	• baby is weighed regularly; • growth should follow a percentile/graph/compare with (growth) chart/graph/compare mass to others of the same age; • detect if the baby gains or loses too much mass;		3

Total 13 marks

Type MS · Explain how the placenta, cord and hormones support a fetus

Answer 1 of 4 · Question 8 part b.i · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Explain how the placenta, cord and hormones support a fetus

Answer 2 of 4 · Question 6 part d · 2 marks

2023 JUN · Paper 02 (regional R) · find part d in this mark scheme extract

6 (a)	• fertilisation; • egg and sperm fuse/gametes/join;		2
(b)	Any four from • mitosis; • DNA is replicated; • chromosomes move to the centre of the cell; • chromosomes are separated to opposite poles; • cell divides;	Allow one mark for prophase, metaphase, anaphase, telophase in the correct order Allow zygote/embyro divides	4
(c)	**D;** (the cells are able to differentiate into any type of body cell)		1
(d)	• (corpus luteum) produces progesterone; • to maintain uterus lining;		2
(e)	Any two from: • umbilical cord established; • transports nutrients/named nutrients/oxygen (to fetus/embryo); • removes waste products/named waste product (from fetus/embryo);		2
(f)	development of male sexual characteristics/genitalia;		1
(g)	**C;** (release of oxytocin from the pituitary gland)		1

Total for Question 6 = 13 marks

Type MS · Explain how the placenta, cord and hormones support a fetus

Answer 3 of 4 · Question 6 part e · 2 marks

2023 JUN · Paper 02 (regional R) · find part e in this mark scheme extract

6 (a)	• fertilisation; • egg and sperm fuse/gametes/join;		2
(b)	Any four from • mitosis; • DNA is replicated; • chromosomes move to the centre of the cell; • chromosomes are separated to opposite poles; • cell divides;	Allow one mark for prophase, metaphase, anaphase, telophase in the correct order Allow zygote/embyro divides	4
(c)	**D;** (the cells are able to differentiate into any type of body cell)		1
(d)	• (corpus luteum) produces progesterone; • to maintain uterus lining;		2
(e)	Any two from: • umbilical cord established; • transports nutrients/named nutrients/oxygen (to fetus/embryo); • removes waste products/named waste product (from fetus/embryo);		2
(f)	development of male sexual characteristics/genitalia;		1
(g)	**C;** (release of oxytocin from the pituitary gland)		1

Total for Question 6 = 13 marks

Type MS · Explain how the placenta, cord and hormones support a fetus

Answer 4 of 4 · Question 4 part b.ii · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

4 (a)	• reduce fat/cholesterol intake; • increase exercise;		2
(b) (i)	Any four from • liver breaks down toxins/drugs; • (if not working properly) statins won't be broken down/will accumulate/reduced detoxification; • kidneys excrete waste (products)/named waste products; • (if not working) can't be removed/less waste/named waste filtered/removed; • increased risk of (further) liver/kidney damage;	Ignore references to deamination	4
(ii)	• statins could cross from mother to fetus; • through the placenta; • damage/affect growth/development of fetus/lead to low birth weight of fetus; • could be in mother's milk if breastfeeding;	Allow baby for mp's 1 and 3 Ignore affects fetus	3
(c) (i)	• (16.3 : 8) = 2.0;	Accept 2.0375 Accept 2/2.04	1
(ii)	Any four from • Double the number/more symptoms shown with patients taking statins than those taking no tablets • even on statins symptoms not high; • almost same with placebo/symptoms with placebo similar/just below those on statins; • symptoms still present when taking no statins; • statins not the cause/other causes of symptoms/named cause e.g. age/gender/fitness/lifestyle;		4 exp

Total 14 marks

Type MS · Explain how the placenta, cord and hormones support a fetus

Answer 4 of 4 · Question 4 part b.ii · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks
			2